Count Each Person at Their Changing Pace: Multi-Person Tempo-Adaptive Repetition Counting

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Abstract

DRAFT — RESULTS PENDING

Multi-person repetitive action counting (MRAC) must assign repetitions to the correct person even when several people act asynchronously and each person accelerates, decelerates, pauses, or disappears behind an occluder. Existing multi-instance systems jointly learn detection, association, and counting from annotated RGB videos, while label-efficient pose counters are primarily developed for a single dominant actor. We study the complementary setting in which a counting model receives identity-indexed pose tracks but uses no person-wise count or period annotation during its intended counting training. Our proposed design shares one pose encoder and a bank of tempo experts across people while keeping temporal state separate for every identity. Within overlapping windows, masked period evidence drives a soft fast–medium–slow router. Expert responses are fused by normalized overlap-add, after which each identity is decoded once, avoiding the duplicate counts produced by summing independent windows. The primary output is a person-wise count vector rather than only a scene total. Experimental evidence is pending synchronization with the collaborator's frozen implementation and audited MultiRep runs.The current manuscript is an evidence-gated pre-results specification: exact auxiliary losses, frontend choices, and every empirical statement remain blocked until they are reconciled with frozen code and machine-readable artifacts.

1. Introduction

Repetitive motion is a useful measurement primitive. Exercise coaching, rehabilitation, industrial monitoring, sports analysis, and human–robot interaction all require more than recognizing that an action repeats: a system must determine how often each participant repeats it. In an uncontrolled scene, however, people need not move together. One person may slow down while another accelerates, a third may pause, and tracks may contain gaps caused by occlusion. A single response curve for the whole frame can therefore mix identities and incompatible time scales, producing a plausible scene total while assigning the wrong count to each person.

Repetition counting has progressed from online local cycle estimation and motion-frequency analysis [4, 8] to learned temporal self-similarity, density regression, and dynamic-query models [1, 3, 6]. Recent work further addresses exemplar conditioning and multiple action types [9, 15]. These advances largely assume one counted stream. MultiCounter formalizes multi-instance repetitive action counting and jointly predicts people, identities, periodic intervals, and counts [10]; MultiCounter+ strengthens temporal identity consistency and long/short-period awareness [11]. Consequently, multi-person, person-wise, asynchronous, and variable-speed counting are established problems rather than novelty claims of this paper.

The remaining opportunity lies at the intersection of supervision, modality, and temporal granularity. Existing MRAC systems are trained as supervised RGB pipelines, whereas pose-based and label-efficient counting has been developed mostly for a single instance. PoseRAC exploits salient poses in an open-set, zero-shot framework [13]; D^2-STX combines RGB and pose streams [12]; and PAMS learns pose representations through period-adaptive multi-scale consistency without repetition annotations [2]. PAMS also evaluates local time warps, but it does not separate simultaneous identities. Conversely, MRAC identity mechanisms do not directly provide an annotation-efficient, pose-driven counter whose tempo assignment can change from window to window along one person's track.

We study that constrained setting. An upstream perception system supplies identity-indexed pose tracks and validity masks. The counting model shares its encoder and tempo experts across people, but maintains independent buffers, routing weights, and output streams for each identity. A masked local period estimate selects a soft mixture of fast, medium, and slow experts within every overlapping window. Instead of counting windows and adding their integer outputs, the model stitches continuous responses with normalized overlap-add and performs one final peak extraction per track. This ordering makes the window boundary explicit and prevents one repetition from being independently counted in two overlapping windows.

Figure 1 contrasts the target output with a mixed single-stream counter. The formulation does not assume that tracking errors have been solved: predicted-track experiments must expose their effect, while oracle-track experiments isolate the counting module. It also does not call the whole system free of all labels. A detector, pose estimator, tracker, or backbone may have external supervision; the narrower intended property is that the counting objective consumes no person-wise repetition count or period labels.

The paper makes three falsifiable, currently evidence-gated contributions:

- We formulate a pose-driven MRAC training setting whose intended counting loss uses no person-wise count or period annotations, while explicitly accounting for externally supervised perception components. This supervision statement must be verified against the frozen training graph before submission.
- We propose identity-conditioned, window-level tempo routing with shared parameters and per-identity state, followed by response-domain overlap-add and one track-level decoding pass. Its benefit must be separated from the effects of tracking and the PAMS representation through controlled ablations.
- We establish an evaluation contract that pairs the official MultiRep counting metrics with oracle/predicted-track protocols, tracking diagnostics, non-stationary-tempo stress tests, and person-count scaling. Every final table cell must be generated from a frozen result manifest.

This pre-results draft freezes the scientific story and its evidence requirements; it does not assert that the multi-person modules are already implemented or that the proposed method outperforms any baseline.

2. Related Work

Single-instance counting and tempo modeling. Early learned counters estimated a local cycle length in an online window, while motion-centric methods used time-varying flow differentials and wavelets to address non-stationary repetition [4, 8]. RepNet made temporal self-similarity a learned bottleneck and introduced Countix for class-agnostic counting in the wild [1]. TransRAC regresses a density map from multi-scale temporal correlations [3], and DeTRC represents cycles with dynamic action queries under a single-primary-action assumption [6]. ESCounts conditions on exemplars and also learns an exemplar-free representation [9]; SkimFocusNet combines coarse context with fine-grained views and studies exemplar-selected multiple action types [15]. The latter task and its Multi-RepCount benchmark differ from MultiRep: they select an action type within a video rather than assign counts to simultaneous people.

Pose-based and label-efficient counting. Pose representations reduce appearance dependence and expose articulated motion directly. PoseRAC selects salient poses and combines foundation-model and text representations for open-set, zero-shot counting [13]; zero-shot prediction should not be conflated with a self-supervised training objective. D^2-STX instead fuses RGB and pose through decoupled spatial and temporal cross-attention [12]. PAMS learns from unlabeled skeleton sequences using period-adaptive multi-scale temporal consistency and combines inference-time counting experts [2]. Its expert consensus is not a learned fast–medium–slow mixture, and its single-stream formulation does not preserve multiple identities. Our intended use of pose and PAMS-TCC is therefore a foundation, not a novelty claim; the proposed distinction is identity-conditioned local routing in MRAC under a no-count-label objective.

Multi-instance repetition counting. MultiCounter defines MRAC and introduces MultiRep, learning detection, association, period localization, and person-wise counting jointly [10]. MultiCounter+ improves robustness and efficiency with a dynamically updated Siamese query and anchor-guided similarity reconstruction that represents long and short periods [11]. These systems already address asynchronous people, variable speeds, pauses, and identity continuity. Our proposed setting differs along a narrower axis: an upstream tracker supplies masked pose tracks, the counting objective is intended to avoid person-wise count and period labels, and tempo specialization varies across local windows of each identity. This modular design is not described as end-to-end, and any comparison to native MRAC systems must disclose the different input and supervision protocols.

3. Method

Status boundary. This section is a provisional method contract for synchronizing the paper with the collaborator's frozen implementation. It defines the intended data flow and the boundary between upstream perception and counting. Components marked **SYNC-REQUIRED** are not asserted as implemented or validated. If the frozen code differs from this contract, the equations and prose must change to match the code.

3.1. Problem Formulation and Overview

Given a video, an upstream detector, pose estimator, and multi-object tracker produce N identity-indexed pose tracks. Track i consists of joint coordinates and a validity mask

$$\mathbf{X}_i = [\mathbf{x}_{i1}, \ldots, \mathbf{x}_{iT_i}] \in \mathbb{R}^{T_i \times J \times C}$$
$$\mathbf{m}_i = [m_{i1}, \ldots, m_{iT_i}] \in \{0, 1\}^{T_i} \quad (1)$$

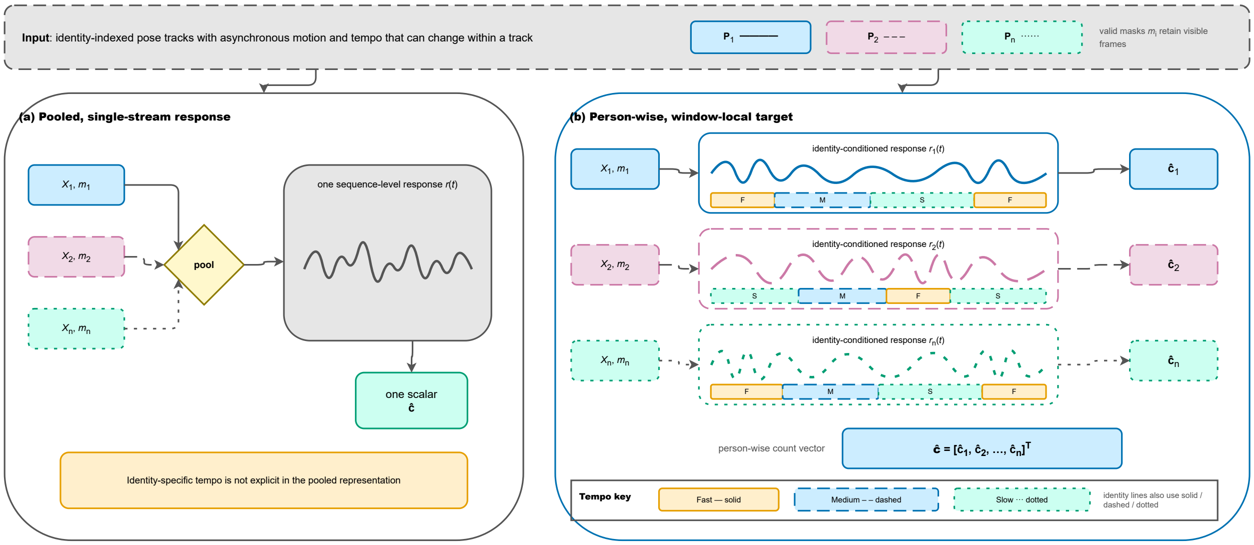

Figure 1. **Target task and output.** A frame-level response can mix people who move at different and changing rates (left). We instead retain identity-indexed masked pose tracks, estimate tempo locally within each track, and return a person-wise count vector (right). The scene total is only a derived diagnostic. This is a problem illustration, not result evidence.

Here T_i is the temporal extent, J the joint count, C the coordinate dimension, and $m_{it} = 1$ denotes a usable pose at time t. Missing observations keep their original time indices and are masked rather than deleted. The primary prediction is

$$\hat{\mathbf{c}} = [\hat{c}_1, \ldots, \hat{c}_N]^\top \in \mathbb{N}_0^N \tag{2}$$

The optional total $\sum_i \hat{c}_i$ is a diagnostic rather than the task definition. The counter cannot undo an identity switch already embedded in its input; Section 4 therefore separates oracle and predicted tracks.

The design in Figure 2 uses a shared pose encoder and a shared bank of fast, medium, and slow experts. "Person-specific" means that observations, masks, temporal buffers, routing variables, and outputs are indexed by identity; it does not mean that a separate parameter set is instantiated for every person. Each track is processed in overlapping windows, whose continuous responses are fused before any discrete counting decision.

3.2. Identity-Conditioned Local Encoding

A single track can accelerate, decelerate, pause, or contain observation gaps, so we do not assign one speed to the whole person. Let L and S denote window length and stride, and let $s_{i\ell}$ be the start of window ℓ. Its local input at $u \in \{1, \ldots, L\}$ is

$$\mathbf{x}_{i\ell u} = \mathbf{x}_{i, s_{i\ell}+u-1}, \qquad m_{i\ell u} = m_{i, s_{i\ell}+u-1} \tag{3}$$

Out-of-range positions are padded with zero mask values. A shared encoder E_θ maps each masked window to embeddings

$$\mathbf{Z}_{i\ell} = E_\theta(\mathbf{X}_{i\ell}, \mathbf{m}_{i\ell}) = [\mathbf{z}_{i\ell 1}, \ldots, \mathbf{z}_{i\ell L}] \in \mathbb{R}^{L \times d} \tag{4}$$

Invalid positions are excluded from aggregation, period evidence, and counting responses. Independent application of the same encoder places people in one representation space without pooling their temporal state.

3.3. Local Period Evidence and Soft Routing

We derive routing evidence within each window. Let the valid-frame mean be $\bar{\mathbf{z}}_{i\ell}$, and define a mask-normalized, non-circular autocorrelation at lag δ

$$A_{i\ell}(\delta) = \frac{\sum_{u=1}^{L-\delta} m_{i\ell u} m_{i\ell, u+\delta} \langle \mathbf{z}_{i\ell u} - \bar{\mathbf{z}}_{i\ell}, \mathbf{z}_{i\ell, u+\delta} - \bar{\mathbf{z}}_{i\ell} \rangle}{\sum_{u=1}^{L-\delta} m_{i\ell u} m_{i\ell, u+\delta} + \epsilon} \tag{5}$$

For a deterministic taper h and an admissible frequency set $\mathcal{F}$, the local spectrum, dominant period, and confidence are

$$S_{i\ell}(f) = |\mathrm{FFT}(h \odot A_{i\ell})_f|^2$$
$$f^*_{i\ell} = \arg\max_{f \in \mathcal{F}} S_{i\ell}(f), \qquad \hat{p}_{i\ell} = 1/f^*_{i\ell} \tag{6}$$

$$\gamma_{i\ell} = \frac{S_{i\ell}(f^*_{i\ell})}{\sum_{f \in \mathcal{F}} S_{i\ell}(f) + \epsilon} \tag{7}$$

The contract uses stopped-gradient embeddings for this evidence so that the encoder cannot reduce its training loss merely by manipulating the routing decision. A router maps the descriptor $\mathbf{q}_{i\ell} = [\log \hat{p}_{i\ell}, \gamma_{i\ell}]^\top$ to a probability simplex

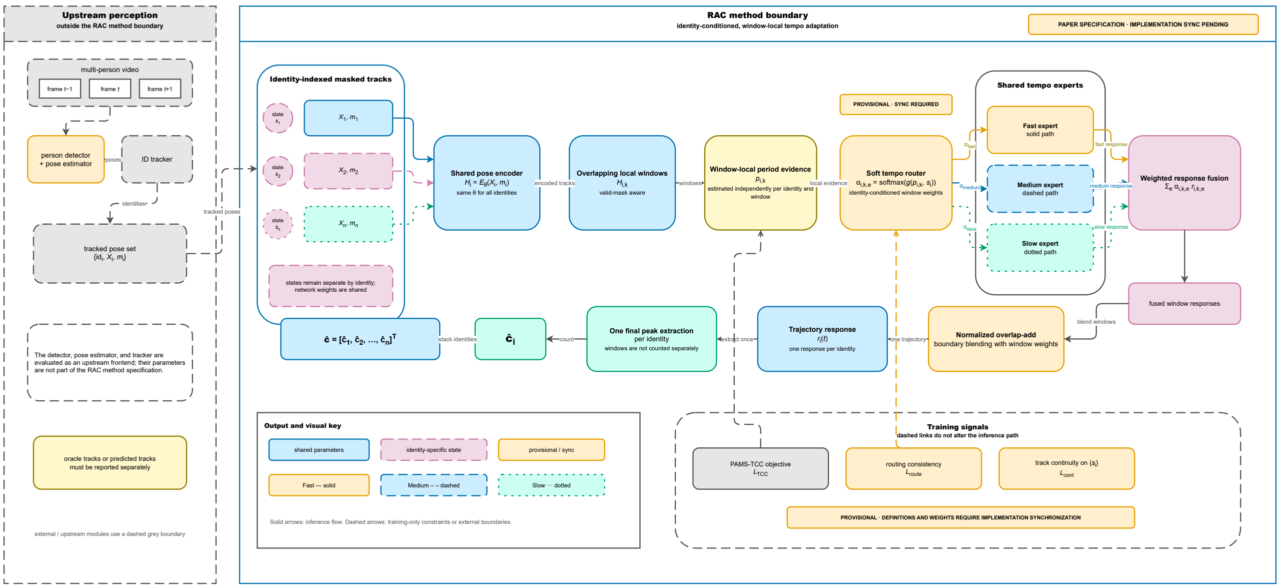

Figure 2. **Provisional method contract.** The upstream detector, pose estimator, and tracker are outside the counting boundary. Within it, parameters are shared across identities while masks, state, local tempo, and output responses remain identity-indexed. Window responses are fused by normalized overlap-add before one final decoding pass per track. Dashed elements require synchronization with frozen training code.

over shared experts $\mathcal{K} = \{\mathrm{f}, \mathrm{m}, \mathrm{s}\}$

$$\mathbf{g}_{i\ell} = \mathrm{softmax}(R_\psi(\mathbf{q}_{i\ell})), \qquad \sum_{k \in \mathcal{K}} g_{i\ell k} = 1 \tag{8}$$

Because $\mathbf{g}_{i\ell}$ is local, it may change between adjacent windows of one identity. Soft assignment also lets an ambiguous transition combine neighboring experts rather than commit to a hard tempo bin.

3.4. Expert Fusion and Boundary Handling

Each expert F_{ϕ_k} predicts a nonnegative response over one masked window. Parameters ϕ_k are shared across identities, while each response remains identity-indexed

$$\mathbf{r}_{i\ell}^{(k)} = F_{\phi_k}(\mathbf{Z}_{i\ell}, \mathbf{m}_{i\ell}, \hat{p}_{i\ell}) \in \mathbb{R}_{\geq 0}^{L}, \qquad r_{i\ell u} = m_{i\ell u} \sum_{k \in \mathcal{K}} g_{i\ell k} r_{i\ell u}^{(k)} \tag{9}$$

Summing integer window counts would double-count cycles that straddle an overlap. We instead reconstruct a single response for each identity. With a nonnegative taper a_u and $\Lambda_i(t) = \{(\ell, u) : t = s_{i\ell} + u - 1\}$, normalized overlap-add gives

$$R_{it} = m_{it} \frac{\sum_{(\ell,u) \in \Lambda_i(t)} a_u r_{i\ell u}}{\sum_{(\ell,u) \in \Lambda_i(t)} a_u m_{i\ell u} + \epsilon} \tag{10}$$

The normalization removes dependence on the number of windows covering an interior frame and excludes padding from the denominator. Only then does a track-level decoder $\mathcal{D}$ extract repetition events

$$\widehat{\mathcal{P}}_i = \mathcal{D}(\mathbf{R}_i, \mathbf{m}_i, \{\hat{p}_{i\ell}, \gamma_{i\ell}\}_\ell), \qquad \hat{c}_i = |\widehat{\mathcal{P}}_i| \tag{11}$$

Thus overlapping windows never contribute separately rounded counts; a cycle near their boundary is resolved once on the fused track response. The frozen implementation must determine the taper, peak threshold, period aggregation, and minimum-distance rule.

3.5. Training Objective and Synchronization Boundary

The intended representation objective builds on PAMS-TCC [2], which uses local and period-offset positives across temporal scales together with within-video, cross-video, and cluster-level negatives. At the method-contract level, training is represented by

$$\mathcal{L} = \mathcal{L}_{\text{PAMS-TCC}} + \lambda_{\text{route}} \mathcal{L}_{\text{route}} + \lambda_{\text{track}} \mathcal{L}_{\text{track}} \tag{12}$$

The reserved routing term should make assignments consistent under count-preserving temporal transforms, while the reserved track term should preserve compatible representations across fragments of one identity. These are responsibilities, not recovered loss definitions.
SYNC-REQUIRED. The exact forms of $\mathcal{L}_{\text{route}}$ and $\mathcal{L}_{\text{track}}$, their weights and sample construction, the pose/tracking

frontend, track normalization, period range, window schedule, router parameterization, expert architectures, and decoder settings must be read from the collaborator's frozen commit and configuration. No contribution or result claim depending on these items is admissible before that synchronization and its corresponding ablation evidence.

4. Experiments

Pre-results status. This section fixes the evaluation questions, protocols, and table schemas; it contains no new-method measurement or empirical conclusion. Red entries are controlled evidence fields. Historical UCFRep dev84 diagnostics, proxy readouts, and runs involving an untrained random Period Head are ineligible for every table below.

4.1. Evaluation Contract

Data and integrity. The primary benchmark will be the exact MultiRep release PENDING under its official PENDING protocol. The original MultiCounter paper and the current MultiCounter+ repository describe different dataset states, so the final artifact manifest must bind the release identifier, download source and time, video/annotation/split SHA-256 digests, evaluator revision, and preprocessing fingerprint [10, 11]. A run is table-ineligible if any binding is missing or if results from incompatible releases are mixed.

Oracle and predicted tracks. The oracle-track protocol supplies official person identities and spatial support, isolating repetition counting from detection and association. The predicted-track protocol uses one frozen detector–pose–tracker frontend for all compatible track-conditioned counters. Native joint MRAC systems retain their audited official pipelines and appear in a separate contextual block, not as if their inputs were controlled. HOTA [5], IDF1, and ID switches [7] diagnose predicted tracks. When the same frozen tracks feed several counters, these tracking values must be identical and cannot be credited as counting gains.

Metrics and uncertainty. Using the frozen official evaluator, we will report Period-mAP, AP50, AP75, AvgMAE, and AvgOBO. Every reproducible method uses at least three pre-registered seeds, reported as mean and sample standard deviation. The primary paired comparison additionally receives a two-sided 95% percentile-bootstrap interval by resampling videos as clusters, preserving all identities within a sampled video. Official single-checkpoint numbers remain explicitly marked and are excluded from variance and significance claims. Test annotations are not used for checkpoint selection, routing calibration, or early stopping.

Baselines and supervision. The mandatory native baselines are MultiCounter and MultiCounter+ [10, 11]. Under the common track interface we adapt RepNet [1], TransRAC [3], PoseRAC [13], and PAMS [2]; DeTRC [6] and D^2-STX [12] enter only if their code and output semantics pass the experiment audit. Tables disclose input modality, training labels, track source, and run count. Comparisons across supervision regimes are accuracy–annotation trade-offs, not claims of identical training information.

4.2. Primary Questions

The evaluation is organized around four questions. First, does a no-count-label pose counter remain competitive on person-wise MRAC under both oracle and predicted tracks? Second, does local routing help beyond a track-conditioned PAMS base when tempo changes within one identity? Third, does response-domain overlap-add reduce boundary errors relative to summing independently decoded windows? Fourth, how do tracking errors, active-person count, and upstream preprocessing affect robustness and efficiency? Each question has a named table or figure and an associated claim that remains withheld until the evidence audit passes.

RESULT-DEPENDENT STATEMENT WITHHELD.

RESULT-DEPENDENT STATEMENT WITHHELD.

The appendix pre-registers router variants, expert counts, local-versus-global tempo, shared-versus-per-person parameters, boundary and continuity ablations, tempo/occlusion/identity-switch stress, cross-dataset transfer, and latency, memory, FLOPs, and throughput as the number of active identities grows. All plots and tables must be generated from frozen JSON, CSV, or Parquet records; manually transcribed values are inadmissible.

5. Conclusion and Limitations

We specify a pose-driven MRAC counter for people whose pace changes within a track. The central design separates identity state while sharing parameters, routes overlapping windows from local period evidence, fuses continuous responses by normalized overlap-add, and decodes each track once. The setting does not remove all annotation and does not jointly optimize perception and counting: the counting objective is intended to avoid person-wise count and period labels, while upstream perception may use external supervision.

This draft has three material limitations. The current local checkout contains only a single-person PAMS reproduction; the multi-person data path, router, continuity loss, and boundary decoder await synchronization with collaborator code. No eligible MultiRep artifact has been received, so the manuscript makes no performance claim. Finally, a modular pose/track pipeline inherits missed detections and identity switches and may scale with the number of active people. The final study must quantify each limitation through

Table 1. Planned MultiRep main results. Reproducible rows will show the mean and sample standard deviation over pre-registered seeds; official single-checkpoint rows will be marked. Protocol blocks answer different questions and will not be pooled into one ranking.

Method	Track protocol	Count supervision	P-mAP↑	AP50↑	AP75↑	AvgMAE↓	AvgOBO↑
RepNet	Oracle	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING
PoseRAC	Oracle	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING
PAMS	Oracle	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING
Ours	Oracle	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING
PAMS	Common predicted	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING
Ours	Common predicted	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING
MultiCounter	Native	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING
MultiCounter+	Native	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING

Table 2. Planned additive ablation under oracle tracks. LT: local tempo; SR: soft routing; BS: boundary stitching; TC: track-continuity objective.

Variant	LT	SR	BS	TC	P-mAP↑	MAE↓
Track-PAMS					PENDING	PENDING
+ LT	✓				PENDING	PENDING
+ SR	✓	✓			PENDING	PENDING
+ BS	✓	✓	✓		PENDING	PENDING
+ TC	✓	✓	✓	✓	PENDING	PENDING

oracle/predicted tracks, corruption analysis, and population-scaling measurements. Until those gates close, the paper remains provisional and data-pending.

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A. Extended Experimental Contract

This appendix records analyses that are required for the final evidence bundle but may move between main paper and supplementary material after results and the official CVPR 2027 format are frozen.

A.1. Artifact Manifest

Every table-eligible run must provide: model commit; evaluator commit; configuration and preprocessing fingerprints; seed; hardware and software environment; MultiRep release, source, download time, split, and checksums; per-video and per-identity ground truth and predictions; oracle/predicted-track status; logs; and generated-table provenance. An artifact lacking any required field is retained for diagnosis but cannot support a paper claim. UCFRep's sealed test partition remains untouched until the method, checkpoints, and analysis code are frozen.

A.2. Mechanism Ablations

The mechanism study compares one global tempo per track against window-local tempo; uniform, hard, and soft routing; one, two, three, and five experts; shared and per-person weights; response fusion and integer-window summation; and training with or without the synchronized continuity term. An oracle router based on ground-truth periods may serve as an upper-bound diagnostic only when the required annotation exists, and must never be grouped with deployable methods.

A.3. Robustness and Tracking Diagnostics

Natural tempo variation will be stratified by within-track cycle-duration dispersion. Deterministic count-preserving piecewise time warps, contiguous masking, and injected identity switches will measure degradation from each clean input using identical corruptions across methods. The protocol must record whether an identity switch changes only the track identifier or also the pose content. Results will be stratified by active-person count, asynchronous starts and stops, pauses, occlusion, and track fragmentation.

A.4. Generalization and Efficiency

If protocol-compatible, a model trained on MultiRep will be evaluated without target-domain adaptation on the official RepCount and UCFRep single-person protocols. Dataset-native normalized MAE and OBO will retain their original names rather than being relabeled as MultiRep AvgMAE/AvgOBO. Efficiency will separate detector/pose/tracker preprocessing from counting latency and report parameters, FLOPs, peak memory, latency, and end-to-end throughput as active identities increase. If ByteTrack [14] is selected as the common tracker, its exact code and checkpoint revision will be frozen in the manifest; this mention does not pre-select the frontend.

Qualitative figures must be generated directly from frozen per-frame artifacts and include persistent identity colors, valid masks, local period estimates, routing weights, fused responses, and ground-truth/predicted boundaries. At least one failure case is mandatory; curves may not be manually redrawn.

Table 3. Planned mechanism analysis. Parameter counts expose the cost of expert count and weight sharing.

Factor	Setting	P-mAP↑	AvgMAE↓	AvgOBO↑	Params (M)↓
Tempo scope	Track-global	PENDING	PENDING	PENDING	PENDING
	Window-local	PENDING	PENDING	PENDING	PENDING
Router	Uniform	PENDING	PENDING	PENDING	PENDING
	Hard	PENDING	PENDING	PENDING	PENDING
	Soft	PENDING	PENDING	PENDING	PENDING
Sharing	Per-person weights	PENDING	PENDING	PENDING	PENDING
	Shared weights	PENDING	PENDING	PENDING	PENDING

Table 4. Planned robustness report. Parenthesized values will be paired changes from the corresponding clean input with video-cluster bootstrap intervals.

Condition	Severity	P-mAP↑	AvgMAE↓	AvgOBO↑	IDF1↑
Clean	–	PENDING	PENDING	PENDING	PENDING
Natural tempo change	Low/medium/high	PENDING	PENDING	PENDING	PENDING
Piecewise time warp	PENDING	PENDING	PENDING	PENDING	PENDING
Contiguous occlusion	PENDING	PENDING	PENDING	PENDING	PENDING
Injected ID switches	PENDING	PENDING	PENDING	PENDING	PENDING

Table 5. Planned efficiency report under one frozen hardware and timing protocol.

Method	Params (M)↓	GFLOPs↓	Peak GB↓	Preprocess ms/frame↓	Model ms/frame↓	FPS↑
MultiCounter	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING
MultiCounter+	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING
Ours	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING